

Tsantiri Artemis

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EDUCATION

- SEP 2020 - JUL 2025 **Michigan State University - Facility for Rare Isotope Beams**
Ph.D in Physics
Department of Physics and Astronomy, College of Natural Sciences

Ph.D Thesis: "*Proton-capture cross-section measurements for the astrophysical γ -process: From stable to radioactive ion beams.*"
Supervisor: Professor Artemisia Spyrou
- SEP 2014 - JUN 2020 **National Technical University of Athens**
Diploma (5-year continuous program) in Physics
Department of Physics, School of Applied Mathematical and Physical Sciences,
GPA: 8.23/10, top 5% in senior class, Very Good

Diploma Thesis: "*Measurement of the fission cross section of the $^{232}\text{Th}(n,f)$ reaction with micromegas detectors.*"
Supervisor: Professor Michael Kokkoris

WORK EXPERIENCE

- JUL 2025 - PRESENT **University of Regina**
Postdoctoral Research Fellow at the Department of Physics with Professor Gwen Grinyer
- Support and lead experiments in TRIUMF laboratory with the ACTAR TPC, IRIS and GRIFFIN collaborations.
 - Analyze data from ACTAR TPC experiment. Codes used: ROOT
 - Supervise and mentor graduate students.
 - Designed and taught nuclear astrophysics lectures for masters level nuclear physics course.
- MAY 2021 - JUL 2025 **Michigan State University**
Research Assistant in Experimental Nuclear Astrophysics at the *Facility for Rare Isotope Beams (FRIB)*
- Participated in experiments in various nuclear physics laboratories (FRIB, Argonne National Laboratory, University of Oslo)
 - Led the setup (vacuum components, electronic setup, digital acquisition system) and execution of my thesis experiment in the ReAccelerator area of FRIB for the measurement of the $^{73}\text{As}(p,\gamma)^{74}\text{Se}$ reaction with the Summing NaI (SuN) γ -ray detector, and developed codes for online and offline analysis
 - Performed data analysis of experiments including energy calibrations, simulations, theoretical interpretation and astrophysical impact calculations. Codes used: ROOT, PYTHON, RAINIER, GEANT4 and NuGrid PPN
 - Mentored undergraduate students within the research group and through departmental mentoring programs

WORK EXPERIENCE (CONT.)

MAY 2024	University of Victoria Visiting Research Fellow at the <i>Department of Physics & Astronomy</i> • Monte Carlo impact studies for the astrophysical i-process using the NuGrid code PPN • Study the impact of experimentally constrained neutron capture reaction rate on the final abundances
SEP 2020 - MAY 2021	Michigan State University Teaching Assistant at the <i>Department of Physics & Astronomy</i> • Graded and hosted homework help sessions for Electromagnetism senior physics course with 60+ students • Instructed Optics laboratory
JUN - JUL 2019	National Center for Scientific Research “Demokritos” Intern at the <i>Institute of Nuclear & Particle Physics</i> • Study of the newly installed BGO detector assembly • Review of the corresponding resolution function • Repairs in two BGO-crystal detector preamplifiers • Re-construction of a preamplifier electronic board (using EAGLE design and schematics software)
2016 - 2020	Home tutor (mathematics / physics) for Middle & High School Students

AWARDS, FELLOWSHIPS AND CERTIFICATIONS

2025	<i>Sherwood K. Haynes Award for Outstanding PhD Student</i> - MSU
2024	Graduate Certificate on <i>Instrumentation in High Energy Physics</i> , Awarded by MSU
2024	<i>Second Prize Oral Presentation</i> in Student/Postdoc competition at the 14 th International Conference on Nucleus-Nucleus Collisions (NN2024)
2024	<i>IReNA Visiting Fellowship</i> of \$4000 to visit the University of Victoria, BC, Canada to work with P. Denisenkov on impact studies
2021	<i>Award for best Teaching Assistant</i> in upper level courses voted by students - MSU

TECHNICAL SKILLS

NUCLEAR PHYSICS:	Codes: GEANT4, TALYS, RAINIER, LISE++, SRIM, SPECTCL, Oslo analysis software Experimental: Beam-line setup, vacuum components, data acquisition systems, soldering, total absorption spectroscopy, NaI detectors, time projection chambers, silicon detectors
NUCLEAR ASTROPHYSICS:	Network Calculations using NUGRID PPN, MESA, NUCNET TOOLS, WEB-NUCLEO
PROGRAMMING & DATA ANALYSIS:	UNIX SCRIPTING, ROOT, C++, PYTHON

OUTREACH-LEADERSHIP-COMMITTEE EXPERIENCE

APR 2026 - PRESENT	<i>Postdoc Representative on CeNAM Executive Committee</i>
NOV 2025 - PRESENT	<i>Membership Administrator for the NuGrid Collaboration</i>
SEP 2025 - APR 2026	<i>Organizing Committee of IReNA's Online Seminar Series</i>
JUN 2024 - OCT 2025	<i>Executive Committee Member of IReNA Blog</i>
MAR 2024 - OCT 2025	<i>Student's Representative on IReNA Steering Committee</i>
SEP 2021 - JUL 2025	<i>Tour guide at the Facility for Rare Isotope Beams (FRIB)</i>
AUG 2022 - JUN 2025	<i>Webmaster for Graduate Student Organization Physics Graduate Organization (PGO) - MSU</i>
JAN 2025 - MAY 2025	<i>Advisory Committee of Open Questions and Research Tools in Nuclear Astrophysics: A Combined Summer School and Hackathon Event - Central Michigan University</i>
AUG 2022 - JUL 2024	<i>Webmaster for Graduate Student Organization Facility for Rare Isotope Beams Graduate Organization (FRIBGO) - FRIB MSU</i>
NOV 2023	<i>CEU Mentor at 2023 Fall Meeting of the APS Division of Nuclear Physics and the Physical Society of Japan</i>
JUN 2023	<i>Activity Leader at NS³ Nuclear Science Summer School 2023 - MSU</i>
MAY 2023	<i>Co-Chair of Organizing Committee of CeNAM Frontiers in Nuclear Astrophysics 2023 Meeting - FRIB MSU</i>
MAY 2023	<i>Organizer of Public Speaking Workshop CeNAM Frontiers in Nuclear Astrophysics 2023 Meeting - FRIB MSU</i>
SEP 2022 - AUG 2023	<i>Mentor undergraduate students through mentoring program held by the Women and Minorities in Physical Sciences (WaMPS) - MSU</i>
AUG 2022 - JUL 2023	<i>Space Committee for Graduate Student Organization Facility for Rare Isotope Beams Graduate Organization (FRIBGO) - FRIB MSU</i>
MAY 2022	<i>Activity Leader at NS³ Nuclear Science Summer School 2022 - MSU</i>
APR 2022	<i>Volunteer at FRIB Countdown Event - FRIB MSU</i>
AUG 2021 - JUL 2022	<i>Webmaster for Graduate Student Organization Women and Minorities in Physical Sciences (WaMPS) - MSU</i>

LANGUAGES

ENGLISH:	Fluent - TOEFL (09/21/2019): 113/120
FRENCH:	Very Good - <i>Certificat Pratique de Langue Française</i> - Paris Sorbonne B2 (07/08/2011)
GREEK:	Native speaker

LECTURES / SEMINARS

JUN 2026 (Upcoming)	<i>Lecture on p-process experiments</i> - 13th European Summer School on Experimental Nuclear Astrophysics, Catania, Italy
MAR 2026	<i>"Introduction to Nuclear Astrophysics"</i> series of three lectures for graduate level nuclear physics course, University of Regina, Regina, SK, Canada
AUG 2025	Physics Department Seminar - University of Regina, Regina, SK, Canada

INVITED CONFERENCE PRESENTATIONS

SEP 2026 (Upcoming)	<i>Nuclear Physics in Astrophysics XII (NPAXII)</i> , Cluj-Napoca, Romania
MAY 2026 (Upcoming)	<i>ECT* Workshop: Nuclear theory for astrophysics: Bridging bound-state and continuum physics</i> , Trento, Italy
JUN 2025	<i>BRIDGCE-IReNA 2025 Annual Meeting</i> , University of York, UK

CONTRIBUTED CONFERENCE PRESENTATIONS

AUG 2026 (Upcoming)	<i>Nuclear Structure 2026</i> , Vancouver, BC, Canada
FEB 2026	(Poster) <i>WNPPC2026</i> , Banff, Alberta, Canada
MAR 2026	<i>AstroNuc2026</i> , Tucson, Arizona, USA
OCT 2024	<i>8th p-process Workshop</i> , Budapest, Hungary
SEP 2024	(Poster) <i>Nuclear Physics in Astrophysics XI (NPA-XI)</i> , Dresden, Germany
AUG 2024	<i>14th International Conference on Nucleus-Nucleus Collisions (NN2024)</i> , Whistler, BC, Canada
JUN 2024	<i>2024 CeNAM Frontiers in Nuclear Astrophysics Meeting</i> , University of Notre Dame, USA
DEC 2023	<i>2023 Fall Meeting of the APS Division of Nuclear Physics and the Physical Society of Japan</i> , Waikoloa Village, Hawaii
NOV 2023	<i>INPART TALYS School</i> , Kruger, South Africa
MAR 2023	<i>18th Russbach School on Nuclear Astrophysics</i> , Rußbach, Austria
OCT 2022	<i>2022 Fall Meeting of the APS Division of Nuclear Physics</i> , New Orleans, USA
MAY 2022	(Poster) <i>JINA-CEE Frontiers in Nuclear Astrophysics Meeting</i> , Notre Dame, USA
MAY 2022	<i>Junior Researcher Workshop - JINA-CEE Frontiers in Nuclear Astrophysics Meeting</i> , Notre Dame, USA
MAY 2022	(Poster) <i>8th Workshop on Level Density and Gamma Strength</i> , Oslo, Norway

LIST OF PUBLICATIONS

- [1] *Constraining the Synthesis of the Lightest p Nucleus ^{74}Se*
Phys. Rev. Lett. 135, 212701, (2025) DOI: [10.1103/d7dr-h36j](https://doi.org/10.1103/d7dr-h36j)
[A. Tsantiri](#), A. Spyrou, E. C. Good, K. Bosmpotinis, P. Giuliani, et al
- [2] *Cross-section measurement of the $^{82}\text{Kr}(p, \gamma)^{83}\text{Rb}$ reaction in inverse kinematics*
Phys. Rev. C 107, 035808 (2023), DOI: [10.1103/PhysRevC.107.035808](https://doi.org/10.1103/PhysRevC.107.035808)
[A. Tsantiri](#), A. Palmisano-Kyle, A. Spyrou, P. Mohr, et al
- [3] *Beta-Decay Half-Lives beyond ^{54}Ca : A Systematic Survey of Decay Properties Approaching the Neutron Dripline*
Phys. Rev. Lett. 136, 092502, (2026) DOI: [10.1103/5yrq-s3g5](https://doi.org/10.1103/5yrq-s3g5)
W.-J. Ong, Z. Y. Xu, R. Grzywacz, A. Ravlic, I. Cox, J.M. Allmond, T.T. King5, B.C. Rasco5, K.P. Rykaczewski, et al
- [4] *Probing the Limits of Statistical Neutron Capture for the r process: Experimental Constraints on ^{141}Cs Nuclear Level Densities*
Phys. Lett. B, 139992, (2025) DOI: [10.1016/j.physletb.2025.139992](https://doi.org/10.1016/j.physletb.2025.139992)
B. Greaves, D. Mcher, A. Spyrou, S. Goriely, D. Rochman, H.C. Berg, D.L. Bleuel, J.A. Clark, C. Dembski, P.A. Deyoung, E. Good, C.M. Harris, A.J. Koning, A.C. Larsen, S.N. Liddick, S.M. Lyons, M. Markova, M.J. Mogannam, G. Owens-Fryar, A. Palmisano-Kyle, A.L. Richard, E.K. Ronning, D. Santiago-Gonzalez, G. Savard, M.K. Smith, A. Sweet, [A. Tsantiri](#)
- [5] *The Upgraded Summing NaI(Tl) (SuN++) Detector*
Nucl. Instrum. Methods Phys. Res. A, 170930, (2025) DOI: [10.1016/j.nima.2025.170930](https://doi.org/10.1016/j.nima.2025.170930)
E. K. Ronning, S. Uthayakumaar, A. Spyrou, A. L. Richard, S. N. Liddick, [A. Tsantiri](#), R. Ringle, et al
- [6] *Universal Effective Charges in the sd and fp Shells*
Phys. Rev. Lett. 135, 072501, (2025) DOI: [10.1103/75ry-71sj](https://doi.org/10.1103/75ry-71sj)
T. H. Ogunbeku, J. M. Allmond, T. J. Gray, W.-J. Ong, B. A. Brown, A. Gargano, R. Grzywacz, J. D. Holt, A. O. Macchiavelli, T. Miyagi, S. Neupane, B. C. Rasco, H. Schatz, B. M. Sherrill, O. B. Tarasov, et al
- [7] *Total Absorption Spectroscopy for the β^+ decay strength distribution of ^{60}Ga*
Phys. Rev. C 111, 065801, (2025) DOI: [10.1103/PhysRevC.111.065801](https://doi.org/10.1103/PhysRevC.111.065801)
G. Owens-Fryar, S. M. Lyons, A. L. Richard, A. Spyrou, B. A. Brown, C. E. P. Robin, Z. Meisel, H. C. Berg, A. Chester, A. A. Chen, B. Crider, P. A. DeYoung, P. Gastis, E. C. Good, C. Harris, K. Hermansen, S. N. Liddick, A. Palmisano-Kyle, A. Psaltis, M. K. Smith, E. Rubino, S. K. Subedi, I. Sultana, and [A. Tsantiri](#)
- [8] *Object Detection with Deep Learning for Rare Event Search in the GADGET II TPC*
Nucl. Instrum. Methods Phys. Res. A, 1080, 170659 (2025) DOI: [10.1016/j.nima.2025.170659](https://doi.org/10.1016/j.nima.2025.170659)
T. Wheeler, S. Ravishankar, C. Wrede, A. Andalib, A. Anthony, Y. Ayyad, B. Jain, A. Jaros, R. Mahajan, L. Schaedig, et al
- [9] *Machine learning enabled measurements of astrophysical (p,n) reactions with the SECAR recoil separator*
Phys. Rev. Research 7, 013074, (2025) DOI: [10.1103/PhysRevResearch.7.013074](https://doi.org/10.1103/PhysRevResearch.7.013074)
P. Tsintari, N. Dimitrakopoulos, R. Garg, K. Hermansen, C. Marshall, F. Montes, G. Perdikakis, H. Schatz, K. Setoodehnia, et al
- [10] *First Study of the $^{139}\text{Ba}(n, \gamma)^{140}\text{Ba}$ Reaction to Constrain the Conditions for the Astrophysical i Process*
Phys. Rev. Lett. 132, 202701, (2024) DOI: [10.1103/PhysRevLett.132.202701](https://doi.org/10.1103/PhysRevLett.132.202701)

- A. Spyrou, D. Mcher, P. A. Denissenkov, F. Herwig, E. C. Good, G. Balk, H. C. Berg, D. L. Bleuel, J. A. Clark, C. Dembski, P. A. DeYoung, B. Greaves, M. Guttormsen, C. Harris, A. C. Larsen, S. N. Liddick, S. Lyons, M. Markova, M. J. Mogannam, S. Nikas, J. Owens-Fryar, A. Palmisano-Kyle, G. Perdikakis, F. Pogliano, M. Quintieri, A. L. Richard, D. Santiago-Gonzalez, G. Savard, M. K. Smith, A. Sweet, [A. Tsantiri](#), and M. Wiedeking
- [11] *Proton Shell Gaps in $N = 28$ Nuclei from the First Complete Spectroscopy Study with FRIB Decay Station Initiator*
Phys. Rev. Lett. 132, 152503, (2024) DOI: [10.1103/PhysRevLett.132.152503](#)
I. Cox, Z. Y. Xu, R. Grzywacz, W.-J. Ong, B. C. Rasco, N. Kitamura, D. Hoskins, S. Neupane, T. J. Ruland, J. M. Allmond, T. T. King, R. S. Lubna, K. P. Rykaczewski, H. Schatz, B. M. Sherrill, O. B. Tarasov, et al
- [12] *Experimentally constrained $^{165,166}\text{Ho}(n, \gamma)$ rates and implications for the s process*
Phys. Rev. C 107, 064614 (2023) DOI: [10.1103/PhysRevC.107.064614](#)
F. Pogliano, A. C. Larsen, S. Goriely, L. Siess, M. Markova, A. Grgen, J. Heines, V. W. Ingeberg, R. G. Kjus, J. E. L. Larsson, K. C. W. Li, E. M. Martinsen, G. J. Owens-Fryar, L. G. Pedersen, S. Siem, G. S. Torvund, and [A. Tsantiri](#)
- [13] *Constraining the astrophysical p process: Cross section measurement of the $^{84}\text{Kr}(p, \gamma)^{85}\text{Rb}$ reaction in inverse kinematics*
Phys. Rev. C 105, 065804 (2022), DOI: [10.1103/PhysRevC.105.065804](#)
A. Palmisano-Kyle, A. Spyrou, P. A. DeYoung, A. Dombos, P. Gastis, O. Olivas-Gomez, C. Harris, S. Liddick, S. M. Lyons, J. Pereira, A. L. Richard, A. Simon, M. K. Smith, [A. Tsantiri](#), and R. Zegers
- [14] *Measurement of the $^{232}\text{Th}(n, f)$ cross section with quasi-monoenergetic neutron beams in the energy range 2–18 MeV*
The European Physical Journal A, 57, 277 (2021), DOI: [10.1140/epja/s10050-021-00590-w](#)
V. Michalopoulou, M. Axiotis, S. Chasapoglou, Z. Eleme, G. Gkatis, A. Kalamara, M. Kokkoris, A. Lagoyannis, N. Patronis, A. Stamatopoulos, [A. Tsantiri](#), and R. Vlastou